INISIATIF KESIHATAN PARU-PARU KEBANGSAAN (LHI) · ALIRAN PESAKIT DARI KLINIK KE HOSPITAL

Integrated Approach to Lung Health Screening and Management

KEMENTERIAN KESIHATAN MALAYSIA
JABATAN KESIHATAN NEGERI KEDAH

PRIMARY CARE · KLINIK KESIHATAN

RED FLAGS
- Unexplained chronic cough
- Stridor
- Unexplained weight loss
- Worsening breathlessness
- Recurrent chest infection
- Frank haemoptysis

Start → Risk Factors for Lung Disease → Symptoms? (Yes) → Red flags? (Yes) → Refer specialist (FMS / Internal Medicine / Respiratory Physician)

Symptoms? (No) / Red flags? (No) → Detailed history, examination and risk assessment

Risk for Asthma

Spirometry → Normal?

Normal? (Yes) → Clinically suspected asthma?
- Clinically suspected asthma? (No) → Follow up as indicated
- Clinically suspected asthma? (Yes) → Elevated biomarkers*?
  - Elevated biomarkers*? (No) → Refer specialist for further evaluation
  - Elevated biomarkers*? (Yes) → Asthma / COPD → Treatment and follow up

Normal? (No) → Obstructive → Asthma / COPD → Treatment and follow up

Normal? → Restrictive → Refer specialist for further evaluation

Risk for COPD

PUMA (≥5) → Spirometry

Risk for Lung Cancer

CXR (with or without AI-assisted) → Normal?

Normal? (Yes) → Low-dose CT
- Lung-RADS 0, 1, 2, 3 → Surveillance
- Lung-RADS 4A, 4B, 4X, S → Refer specialist for further evaluation

Normal? (No) → Nodule / Mass → Refer specialist for further evaluation

Normal? (No) → Consolidation, atelectasis, pleural effusion etc → Refer specialist for further evaluation

Risk for Tuberculosis

- Symptomatic → Follow CPG for TB work-up
- Asymptomatic → Refer: High Risk Group TB Screening Guidelines

SPECIALIST CARE · HOSPITAL

Treatment and follow up
- Difficult to treat → Referral via Severe Asthma Network → MDT → Decision for biologic → Follow up
- Frequent exacerbator → Refer to specialist → MDT → Optimise treatment · Decision for biologic → Follow up

Refer specialist for further evaluation
- CECT / PET scan → Confirmatory investigation → Basic & advanced bronchoscopic procedure / image-guided biopsy / pleuroscopy / surgical resection → Confirmed histopathology → Molecular testing → MDT → Decision for treatment
- Other lung diseases → Investigate and treat as indicated

***Biomarkers:** blood eosinophil count, total and specific IgE, FeNO (fraction of exhaled nitric oxide).

PUMA: COPD case-finding questionnaire; score ≥5 → spirometry.

MDT: multidisciplinary team.

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